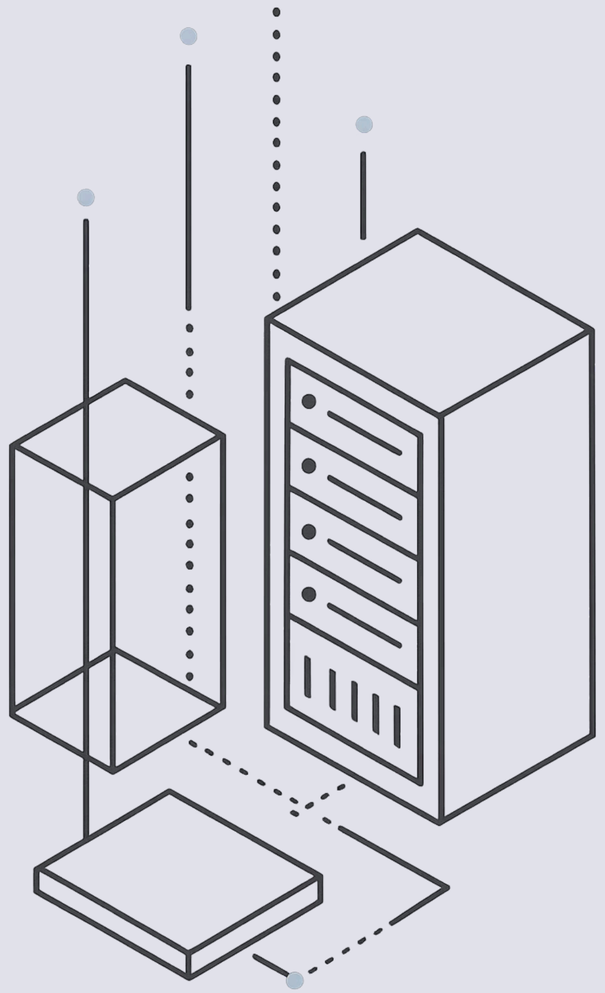
Beyond Basic RAG
Better Context for More Reliable Enterprise AI

Ravi Ray

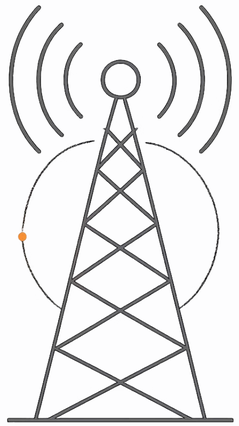
Software Engineering Architect

ENTERPRISE RAG FRAMEWORK

CONF42 PROMPT ENGINEERING 2026



Why Retrieval Quality Outweighs Prompt Quality



In enterprise AI, the **quality of retrieved context** not prompt wording is the primary driver of model accuracy and consistency.

Knowledge Freshness

Stale context produces stale answers.

Information Governance

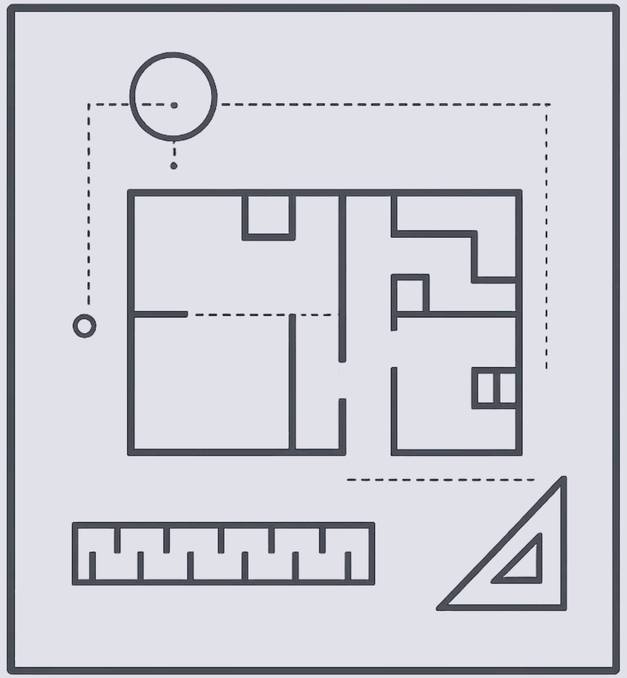
Who sees what matters as much as what is retrieved.

Retrieval Precision

Irrelevant chunks degrade even perfect prompts.

RAG Is Now a Foundation, Not a Feature

Retrieval-Augmented Generation has moved from an experimental pattern to the **core architecture** for enterprise AI. Designing effective retrieval pipelines is no longer optional it is the discipline that separates reliable production systems from brittle prototypes.



THE FRAMEWORK

A Five-Axis Framework for Enterprise RAG



Ingestion

Parsing, chunking, and normalizing diverse enterprise content



Indexing

Dense, sparse, and hybrid index strategies



Retrieval Composition

Query rewriting, re-ranking, and multi-stage pipelines



Grounding

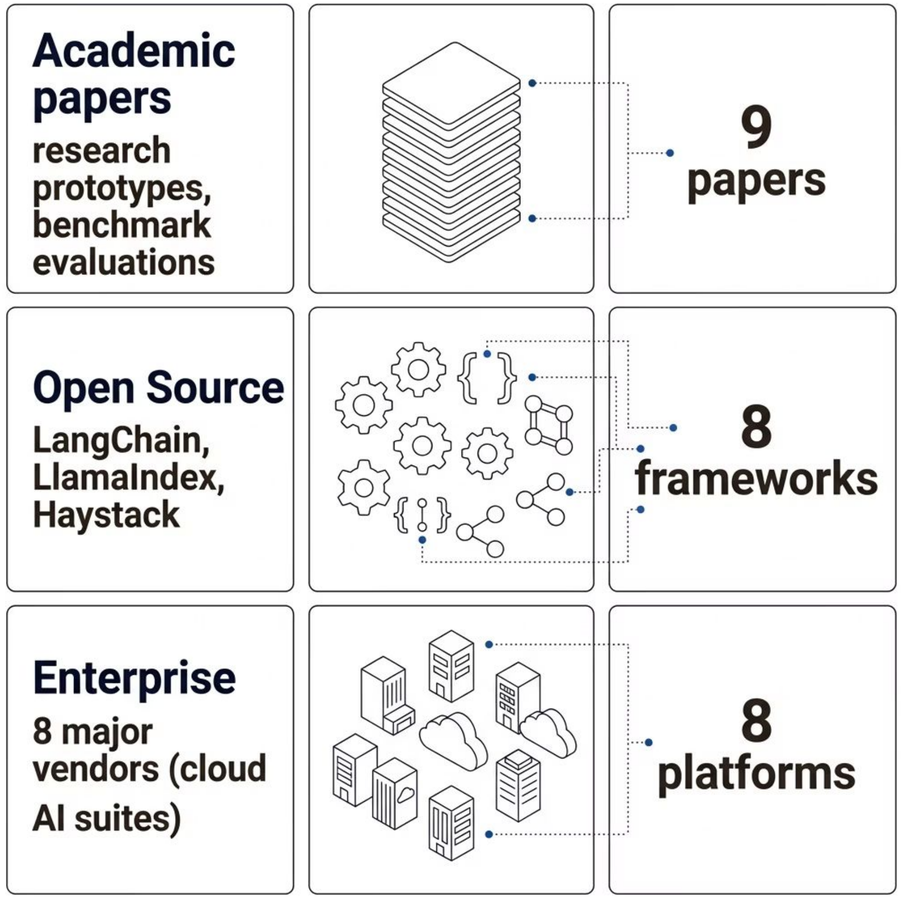
Citation, attribution, and faithfulness to source



Governance

ACL-aware retrieval, provenance, and audit trails

How This Framework Was Built



The five-axis framework is grounded in a **comparative study of ~25 implementations** spanning academia, open source, and commercial enterprise platforms.

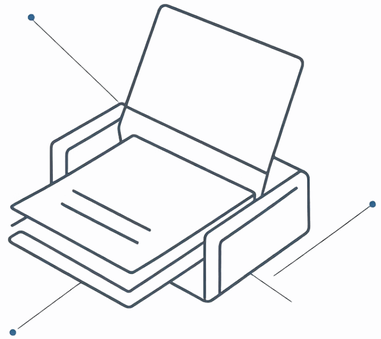
- Examined how production systems deviate from research prototypes
- Identified capabilities that appear only in enterprise deployments
- Synthesized patterns across **8 major enterprise platforms**

This is a vendor-neutral analysis no single framework or vendor is prescribed.

AXIS 1

Ingestion: Where Quality Is Won or Lost

Most retrieval failures originate **before** the vector store. Ingestion pipelines must handle format diversity, extract structure, and chunk content in ways that preserve semantic coherence.



Format Diversity

PDFs, HTML, tables, slides, code

Chunk Strategy

Fixed, semantic, hierarchical, and agentic chunking

Metadata Enrichment

Author, date, access level, lineage

AXIS 2 + 3

Indexing & Retrieval Composition



Dense Retrieval

Embedding-based semantic search. Captures meaning but struggles with exact terms and identifiers.



Sparse Retrieval

BM25 and keyword-based. Precise for known terms but blind to paraphrase and synonymy.



Hybrid Fusion

Combines dense and sparse scores. The standard for enterprise accuracy at scale.



Re-ranking

Cross-encoder re-ranking refines top-k candidates before context is passed to the model.

AXIS 4

Grounding: Making AI Answers Trustworthy



Grounding strategies determine how faithfully a model's response reflects retrieved content and whether that connection is **verifiable**.

01

Attribution

Link every claim to a source chunk and document

02

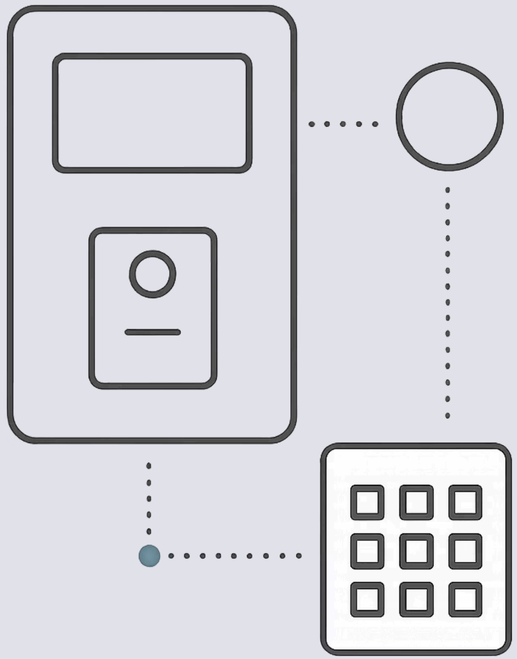
Faithfulness Scoring

Detect when models hallucinate beyond retrieved context

03

Confidence Signaling

Surface retrieval gaps to users rather than fabricating answers



AXIS 5

Governance: Enterprise-Grade Retrieval Controls

Governance capabilities are the sharpest line between research prototypes and production enterprise systems. Three capabilities matter most:

ACL-Aware Retrieval

Enforce document-level permissions at query time not just at the UI layer

Provenance Tracking

Record the full lineage of retrieved chunks for audit, compliance, and debugging

Time-Aware Evaluation

Assess answer correctness relative to document date, not just semantic similarity

7 Open Engineering Challenges

1 **Long-Document Grounding**

Reliably attributing answers across documents exceeding context windows

2 **Heterogeneous Content Retrieval**

Unified ranking across text, tables, code, images, and structured data

3 **Knowledge Freshness via CDC**

Change Data Capture pipelines that keep vector stores synchronized with live systems

4 **Confidentiality-Preserving Evaluation**

Benchmarking retrieval quality without exposing sensitive enterprise content

3 More Challenges Shaping Enterprise AI

Multi-Agent Retrieval

Coordinating retrieval across autonomous agents without context collision, redundancy, or conflicting grounding

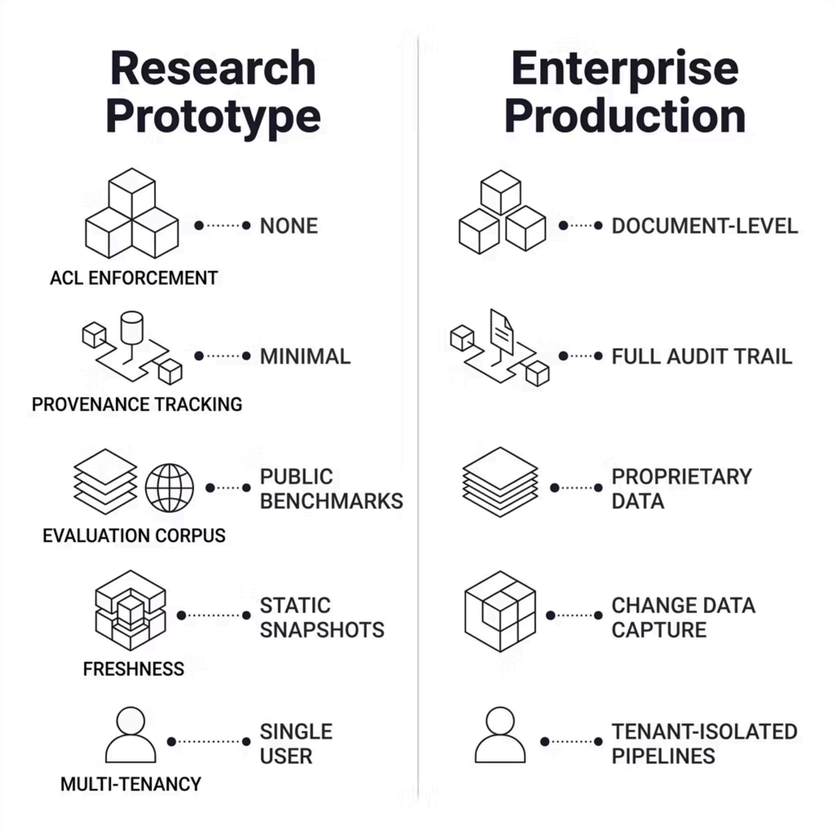
Latency vs. Cost Trade-offs

Re-ranking and multi-stage pipelines improve quality but increase p99 latency and operational cost requiring deliberate optimization

Evaluation at Enterprise Scale

Standard RAG benchmarks do not reflect enterprise conditions: access controls, proprietary corpora, mixed modalities, and SLA constraints

Production vs. Research: Where They Diverge



The gap between research RAG and production RAG is **not primarily algorithmic**. It is operational, organizational, and regulatory.

Enterprise platforms must address constraints that academic benchmarks deliberately exclude access control, auditability, and confidentiality making them fundamentally different engineering problems.



Designing Knowledge Systems, Not Just Pipelines

The highest-leverage shift in enterprise AI is reframing RAG as a **knowledge system design problem** with the same rigor applied to data architecture, schema governance, and access control as any production database.

Key Takeaways

Context Quality First

Retrieval quality has more impact on output reliability than prompt engineering

Five Axes to Evaluate

Ingestion → Indexing → Retrieval Composition → Grounding → Governance

Governance Is Non-Negotiable

ACL-aware retrieval and provenance tracking are production requirements, not nice-to-haves

Vendor-Neutral Lens

Apply this framework across platforms the principles hold regardless of stack

What to Do Next

01

Audit Your Ingestion Pipeline

Identify chunking gaps, missing metadata, and format coverage

02

Implement ACL-Aware Retrieval

Move access control enforcement to the retrieval layer, not the application layer

03

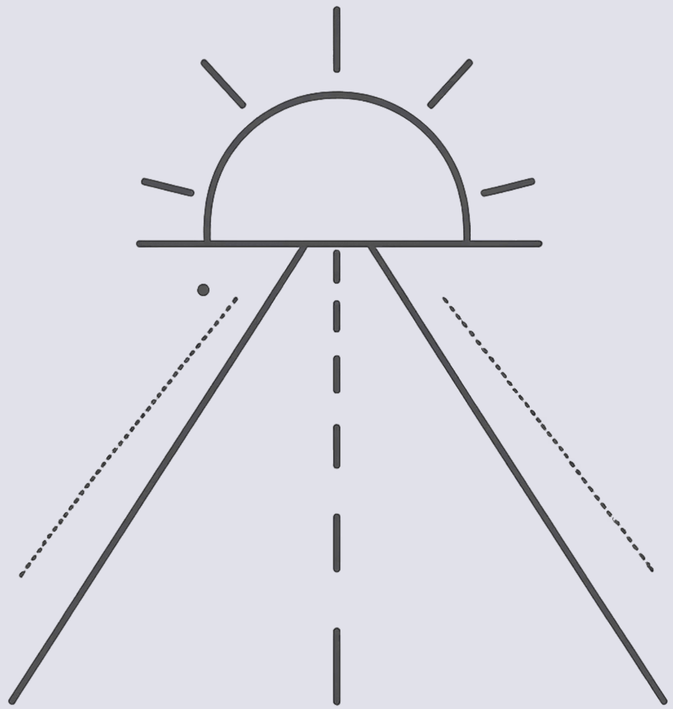
Instrument for Provenance

Log retrieved chunks and scores for every query enable traceability from day one

04

Build an Enterprise Eval Suite

Develop domain-specific, access-controlled benchmarks that reflect your real retrieval conditions



Thank You!

Ravi Ray

Software Engineering Architect

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